Aerospace Math Appendix

Standard SI units. Symbols: L (lift), D (drag), T (thrust), ρ (air density), V (true airspeed), C_L, C_D (coefficients), S (wing area), m (mass), g (9.80665 $m \cdot s^{-2}$), W = mg (weight).

Aerodynamics

$L = \frac{1}{2} \rho V^2 C_L S$ — Lift equals dynamic pressure times lift coefficient and wing planform area.

$D = \frac{1}{2} \rho V^2 C_D S$ — Drag equals dynamic pressure times drag coefficient and wing area.

$q = \frac{1}{2} \rho V^2$ — Dynamic pressure is one half air density times speed squared.

$C_D = C_{D0} + k C_L^2$, $k = 1/(\pi e AR)$ — Parasitic and induced drag combine via a quadratic polar with k set by aspect ratio AR and Oswald efficiency e.

$Re = \rho V c / \mu$ — The Reynolds number uses density, speed, chord c, and dynamic viscosity μ to characterize viscous effects.

$M = V / a$ — The Mach number is speed divided by local speed of sound a.

Definitions: S [m^2], V [$m \cdot s^{-1}$], ρ [$kg \cdot m^{-3}$], μ [Pa·s], a [$m \cdot s^{-1}$], $AR = b^2/S$.

Propulsion

$T \approx D$ (steady, level) — In steady, level flight thrust approximately balances drag.

$P_{req} = D V$ — Propulsive power required equals drag times speed.

$P_{avail} = \eta_p P_{shaft}$ (prop) ; $P_{avail} = T V$ (jet) — Available power is shaft power times propulsive efficiency for propellers, or thrust times speed for jets.

$\eta_{p,ideal} = 2 / (1 + V_j/V)$ — Ideal propulsive efficiency depends on jet slipstream speed V_j relative to flight speed V.

$T = \dot{m} (V_j - V) + (p_e - p_a) A_e$ — General thrust expression combines momentum and pressure terms at the nozzle exit.

$TSFC = \dot{m}_f / T$; $BSFC = \dot{m}_f / P$ — Thrust specific fuel consumption (TSFC) is fuel flow per thrust; brake specific (BSFC) is fuel flow per power.

Units: TSFC c_T [s^{-1}] (mass flow per thrust), BSFC [$kg \cdot W^{-1} \cdot s^{-1}$] (mass flow per power).

Performance

$V_{stall} = \sqrt{2 W / (\rho S C_{L,max})}$ — Stall speed follows from lift equal to weight using maximum lift coefficient.

$(L/D)_{max}$ at $C_L = \sqrt{C_{D0} / k}$ — Maximum aerodynamic efficiency occurs where induced and parasitic drag contributions are equal.

$RC = (P_{avail} - P_{req}) / W$ — Rate of climb equals excess power divided by weight.

$R_{jet} = (V / c_T)\,(L/D)\,\ln(W_i/W_f)$ — Breguet range for jets uses speed V, thrust SFC c_T, and lift-to-drag ratio.

$R_{prop} = (\eta_p / (BSFC \cdot g))\,(L/D)\,\ln(W_i/W_f)$ — Breguet range for propeller aircraft uses propulsive efficiency and mass-based BSFC; divide by g for consistency with weight-based fuel fraction.

$E_{prop} = (\eta_p / (BSFC \cdot g))\,(C_L^{3/2} / C_D)\,\sqrt{2 \rho S / W_{avg}}$ — Prop endurance increases with aerodynamic efficiency and air density and decreases with weight.

$\omega = (g \tan \varphi) / V$; $n = 1 / \cos \varphi$ — Coordinated level turn rate ω ($rad \cdot s^{-1}$) and load factor n depend on bank angle φ and speed V.

> Constants & units: $g = 9.80665\ m \cdot s^{-2}$, W [N], c_T [s^{-1}], BSFC [$kg \cdot W^{-1} \cdot s^{-1}$], RC [$m \cdot s^{-1}$], ω [$rad \cdot s^{-1}$].

Navigation / Stability

$\psi_m = \psi_t + \delta_{var} + \delta_{dev}$ — Magnetic heading equals true heading plus variation and local deviation.

$x_{te} \approx d \sin(\Delta\chi)$ — Cross-track error after along-track distance d is distance times the sine of the course error Δχ (small-angle).

Drift ≈ $\tan|\Delta\psi| \times$ distance — Lateral drift grows linearly with distance at small heading error Δψ (as used in compass-variance estimation).

$C_{n\beta} > 0$; $C_{m\alpha} < 0$ — Directional and longitudinal static stability require positive weathercock derivative and negative pitch stiffness.

$TAS \approx IAS \sqrt{\rho_0/\rho}$ — At subsonic speeds, true airspeed scales with indicated airspeed by the square-root of the density ratio.

V_w crosswind limit: $|\beta| \approx \arcsin(V_x/V)$ — Allowable crosswind component V_x sets a sideslip constraint at approach speed V.